

9.8 Comparing Distributions

Lesson Introduction

 Benchmarks	MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range. MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.		 Learning Targets	- I can find and interpret the mean and median of data sets. - I can use mean and median to compare data sets. - I can compare the distribution of data sets.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.		MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.	
 Essential Questions	- When would it be better to use a line plot over a histogram? - What does the mean tell you about a data set? - What does the distribution tell you about a data set?			
 Lesson Narrative	In this lesson, students will build on their understanding of histograms and line plots. Students will find the mean or median within a data set and compare it to another data set. Students will synthesize this information to use for their reasoning behind their answers. Ensure students have a firm understanding of how to read histograms and line plots before beginning this lesson.			
 Component Summary	9.8.1 Warm-Up (~5 mins)	Engage in “Which one doesn’t belong?” routine (<i>SE p. 84</i>)	9.8.3 Guided Practice (15 mins)	Determine the mean and median in line plots, histograms, and tables (<i>SE p. 85</i>)
	9.8.2 Collaboration (10 mins)	Match labels to line plots (<i>SE p. 84</i>)	9.8.4 Your Turn! (10 mins)	Use mean and median to compare data sets and compare the distribution of data sets (<i>SE p. 87</i>)
	9.8.5 Wrap-Up (~5 mins)	Compare distribution sets of histograms (<i>SE p. 89</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Distribution - Mean and Median - Histogram and Line Plot		(Optional) Paper	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may incorrectly switch the median and the mode. - Student may add numbers that are labels at the bottom instead of the amount in each category. - Students may incorrectly think that the given histograms show exact numbers for the data points.	Make sure students can read each chart and can see all of the labels. Consider projecting each line plot or histogram on the board for those who are visually impaired
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Which One Doesn't Belong? The 9.8.1 Warm-Up engages students in a “Which one doesn't belong?” routine. Use this as an opportunity to encourage critical thinking and reasoning skills. Each of the four options may not belong for some reason. As students discuss their responses, encourage them to justify their thinking using the pictures.	MA.K12.MTR.6.1: Reasonableness Throughout this lesson students make comparisons between data sets before finding measures of central tendency. Students have the opportunity to predict solutions based on the given information, strengthening their ability to reason mathematically about data sets and to consider the reasonableness of their hypotheses.
 Differentiate Instruction	Line plots and histograms provide visual entry points for learners throughout this lesson. Seeing the data displayed graphically can help students make connections and hypothesize about measures of central tendency as well as the distribution and spread of the data, which is critical to being able to compare data sets.	
Lesson Notes:		

9.8.1 Warm-Up: Which One Doesn't Belong?

Component Overview: Students engage in a “Which One Doesn't Belong?” routine to determine the license plate that doesn't belong with the rest.



9.8.1 Warm-Up: Which One Doesn't Belong?

- Which license plate doesn't belong? Explain your reasoning.



Answers will vary.



Which One Doesn't Belong?

Engage students in a “Which one doesn't belong?” routine, prompt students to explain their rationale for their choice, and give opportunities for students to refine and clarify their answers.

9.8.2 Collaboration: Comparing Distributions

Component Overview: Students work with a partner to match line plots to their corresponding grade levels based on summaries describing the data for each class.



9.8.2 Collaboration: Comparing Distributions

- Clare recorded the amount of time students in the sixth, eighth, and tenth grades spent doing homework, in hours per week. She made a line plot of the data for each grade and provided the following summary.
 - Students in sixth grade tend to spend less time on homework than students in eighth and tenth grades.
 - The homework times for the tenth-grade students are more alike than the homework times for the eighth-grade students.

Use Clare's summary to match each line plot to the correct grade (sixth, eighth, or tenth).

Grade Level	Line Plot
<i>eighth grade</i>	
<i>sixth grade</i>	
<i>tenth grade</i>	
	Time (in hours)



Consider encouraging students to label the line plots by circling the median, drawing a line down the center, and comparing both sides to draw conclusions about the data. This will provide visual entry points for students to analyze the data presented in the line plots and allow for easier comparison.



Take Turns

With a partner, students take turns deciding which grade belongs with each graph. When Partner A chooses an answer, Partner B will ask them to give reasons for their answer and then agree or disagree with a reason. Students will discuss until they come to an agreement. Consider switching partners or providing prompting questions if students cannot come to an agreement.

9.8.3 Guided Practice: Comparing Distributions

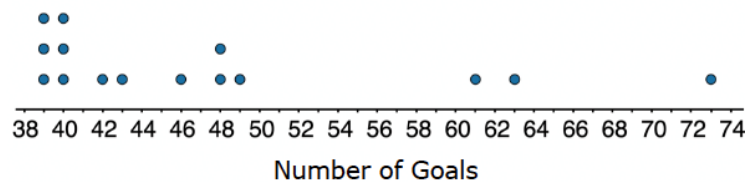
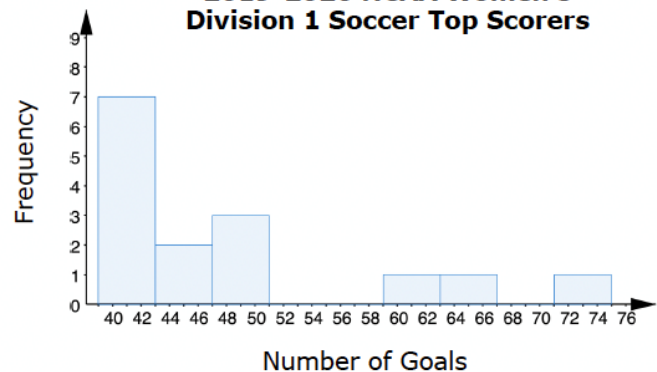
Component Overview: Students analyze a data set presented as a histogram and a line plot and in a table. Students determine which measures of central tendency can be found using a line plot and a histogram and consider which measure of central tendency will be larger given the data set. Students then analyze the same data set presented in a table using only the top ten data points to examine how the measures of central tendency change.



9.8.3 Guided Practice: Comparing Distributions

1. A histogram and line plot of the number of goals scored by the top 15 women goal scorers during the 2019-2020 NCAA Women's Division 1 soccer season are shown.

2019-2020 NCAA Women's Division 1 Soccer Top Scorers



- a. The

☐ histogram
☒ line plot
☐ histogram and line plot

 can be used to determine the mean.
- b. The

☐ histogram
☒ line plot
☐ histogram and line plot

 can be used to determine the median.



Differentiation is built in throughout this component to provide auditory and visual entry points. Graphical displays of data allow students to visualize the center and spread of the data set. Partner discussion provides auditory entry points.



This section is intentionally scaffolded to solidify student understanding of how data impacts measures of central tendency. Students begin by using visual representations of data to generalize about mean and median, then progress to using data presented in a table to find mean and median. Encourage students to discuss how the different graphical displays impact the measures of center that can be calculated.

9.8.3 Guided Practice: Comparing Distributions (continued)

- c. Which measure of central tendency do you expect to be larger, the mean or the median? Explain your reasoning.

Answers will vary. I expect the mean to be larger because the larger values that cause the distribution to be skewed to the right increase the average.

2. The table lists the data for the top 15 women goal scorers during the 2019-2020 NCAA Women's Division 1 soccer season.

Rank	Player	Goals
1	Evelyn Viens, South Fla.	73
2	Catarina Macario, Stanford	63
3	Raimee Sherle, Boise St.	61
4	Ally Watt, Texas A&M	49
5	Tziarra King, NC State	48
5	Deyna Castellanos, Florida St.	48
7	Alexa Genas, Campbell	46
8	Morgan Weaver, Washington St.	43
9	Peyton DePriest, Middle Tenn.	42
10	Elise Flake, BYU	40
10	Samantha Dewey, Xavier	40
10	Hayley Younginer, Wake Forest	40
13	Dakota Mills, Saint Joseph's	39
13	Katie McClure, Kansas	39
13	Taylor Kornieck, Colorado	39

- a. Find the mean number of goals. Interpret the mean.

47.3. The average number of goals scored for the top 15 women during the 2019-2020 NCAA Women's Division 1 soccer season was 47.3 goals.



Students may struggle to predict the measure of center that will be larger. Help students think through and estimate their answers by asking questions like:

- What do you notice about the data? Is there wide variation?*
- Does a wide variation affect mean or median more? Why?*
- Are there any outliers? How do outliers typically affect measures of central tendency?*

9.8.3 Guided Practice: Comparing Distributions (continued)

- b. Find the median number of goals. Interpret the median.

43. The median number of goals scored for the top 15 women during the 2019-2020 NCAA Women's Division 1 soccer season was 43 goals. Half of the top 15 scored fewer goals and half scored more.

- c. Which value is larger, the mean or the median?
mean

- d. How would the mean and median change if you found the mean goals and median goals for the top 8 scorers?

Answers will vary. The mean and the median would both be larger numbers.

- e. Find the mean number of goals for the top 8 scorers.
53.875

- f. Find the median number of goals for the top 8 scorers.
48.5

- g. Discuss with your partner how the measures of central tendency changed. Write your conclusion.
Answers will vary. Both the median and the mean increased since the lowest seven scorers were removed from the data set.



Take Turns

Encourage students to engage in mathematical discourse through a Take Turns routine. Students work in pairs or small groups and take turns in the work of the component. Circulate as students work and encourage partners to question each other's work, justify their own claims, and explain their thinking.



Consider allowing students to explain their answers to questions verbally instead of in writing. Oral explanations may help students who struggle with written response articulate their thinking and provide another valid form of assessment.

9.8.4 Your Turn!

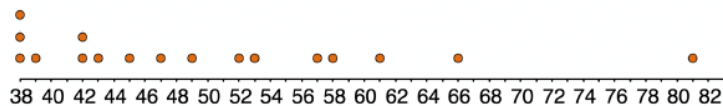
Component Overview: Students compare two data sets displayed on line plots. Students begin by describing the differences between the data sets, then hypothesize about the mean and medians and compare their predictions with a partner. Students then find and interpret the measures of center for one of the data sets by using data displayed in a table.



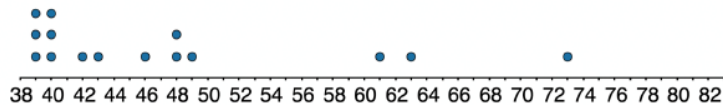
9.8.4 Your Turn!

- The line plots show the top 15 women goal scorers during the 2016-2017 and 2019-2020 NCAA Women's Division 1 soccer season.

2016-2017 Season



2019-2020 Season



Number of Goals

- What is different about the top 15 goal scorers during the 2016-2017 soccer season and the top 15 goal scorers during the 2019-2020 soccer season?

Answers will vary. The number of goals are closer together (less variation) during the 2019-2020 season versus the 2016-2017 season. There is a larger range for the 2016-2017 season.

- Make a hypothesis determining whether the mean of the 2019-2020 season or the mean of the 2016-2017 season is higher. Explain what influenced your hypothesis.

Answers will vary. My hypothesis is that the mean of the 2016-2017 season is higher because that season has more data points with larger values.



Students may struggle to compare the data sets and hypothesize about the measures of center. Provide scaffolding by asking questions like:

- How can you describe the data of the 16-17 season? The 19-20 season?
- Does either season have outliers? How will this impact the data?
- Which measure of center is more impacted by outliers? How can this help you hypothesize about the means and medians of the data sets?

9.8.4 Your Turn! (continued)

- c. Make a hypothesis determining whether the median of the 2019-2020 season or the median of the 2016-2017 season is higher. Explain what influenced your hypothesis.

Answers will vary. My hypothesis is that the median of the 2016-2017 season is higher because most of the values for the 2019-2020 are closer together and are smaller values.

- d. Compare your predictions with your partner's predictions. Make any adjustments needed based on your conversation.
2. The table shows the list of the top 15 women goal scorers during the 2016-2017 NCAA Women's Division 1 soccer season.

Rank	Player	Goals
1	Savannah Jordan, Florida	81
2	Tabitha Tindell, FGCU	66
3	Rachel Hill, UConn	61
4	Mckenzie Meehan, Boston College	58
5	Aaliyah Lewis, Alabama St.	57
6	Tyler Lussi, Princeton	53
7	Valerie Sanderson, Memphis	52
8	Murielle Tiernan, Virginia Tech	49
9	Ashley Hatch, BYU	47
10	Ariela Lewis, Alabama St.	45
11	Emily Gingrich, Saint Joseph's	43
12	Margaret Purce, Harvard	42
12	Jaycie Johnson, Nebraska	42
14	Hayley Dowd, Boston College	39
15	Rachel Holden, North Texas	38
15	Lauren Miller, North Dakota St.	38
15	Abby Reed, DePaul	38

- a. Find the mean number of goals for the 2016-2017 soccer season.

49.9



Encourage students to analyze and respond to their partners' predictions to encourage mathematical discussion. Ensure students are using formal vocabulary, such as variability, spread, symmetry, and distribution to justify their responses.

9.8.4 Your Turn! (continued)

- b. Interpret the mean.

Answers will vary. The average goals scored for the top 15 women during the 2016-2017 NCAA Women's Division 1 soccer season was 49.9 goals.

- c. Find the median number of goals for the 2016-2017 soccer season.

47

- d. Interpret the median.

Answers will vary. The median number of goals scored for the top 15 women during the 2016-2017 NCAA Women's Division 1 soccer season was 47 goals. Half of the top 15 scored fewer than 47 goals and half scored more than 47 goals.

- e. Complete the statements.

The mean of the 2016-2017 soccer season is

- ☐ less
☒ greater

than the mean of the 2019-2020 soccer

season. The value in the ☐ 2016-2017
☐ 2019-2020 soccer

season that likely influenced the mean is 81.

The median of the 2016-2017 soccer season is

- ☐ less
☒ greater

than the median of the 2019-2020

soccer season.



For a project-based learning activity, consider allowing students to research data on a topic they are interested in that has two different sets of data to create tables and line plots for each set of data to compare.

9.8.5 Wrap-Up: Ticket Out the Door

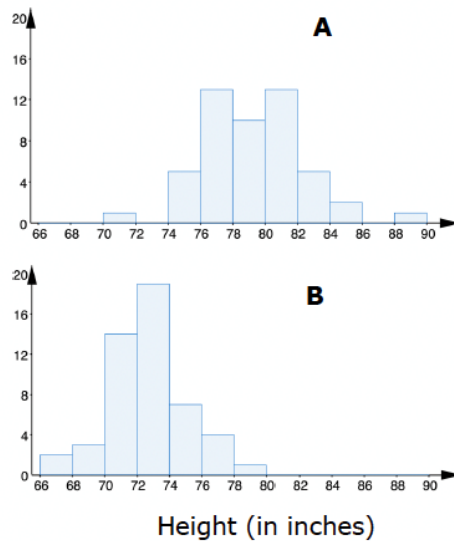
Component Overview: Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson. This Wrap-Up assesses students' abilities to compare distribution in data sets.



9.8.5 Wrap-Up: Ticket Out the Door

1. Professional basketball players tend to be taller than professional baseball players.

The two histograms, labeled A and B, show height distributions of 50 male professional baseball players and 50 male professional basketball players.



Data from this Ticket Out the Door may be useful to inform small groups for remediation or reteaching opportunities.

- a. Choose one the titles: "Heights of Baseball Players" or "Heights of Basketball Players" for each histogram.

Histogram	A	B
Best Title	Heights of Basketball Players	Heights of Baseball Players

- b. Describe the spread and shape of the distribution of the heights of the basketball players.

Answers will vary. The heights of professional basketball players are symmetric, with most players between 76 and 82 inches tall. The spread of the distribution shows a range from 70 inches tall up to 90 inches tall.